X-ray Tube Heater Current Drift Detection Using Uploaded Field Data

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In this final project, data-driven methods to detect heater current drift in the MesoFocus 450 kV (MF-450) X-ray source product line are developed and evaluated, using field data available in Databricks. The issue affects a small subset of MF-450 X-ray tubes at end-user sites. These tubes show an anomalous heater current drift at a given operating point and reduced emission performance, which leads to visible image quality degradation. In addition, operating the tubes at maximum heater current for prolonged periods accelerates tube wear and shortens lifetime, causing unscheduled downtime and measurable costs. The resulting customer dissatisfaction creates operational workload for internal support teams and introduces commercial risk, including reduced reputation in the worst case. From a data science perspective, the problem is challenging because only a few tubes have been officially observed with this behavior, which makes standard supervised classification difficult. To overcome this, individual heater current time-series samples are treated as data points, enabling statistical modeling while still reflecting imbalance.

Building a practical detection model with three priorities is targeted: first, maximize recall to avoid missing drifting tubes; second, control precision to reduce false alarms and associated operational effort; and third, maximize the lead time between drift detection and tube failure. After exploratory data analysis with descriptive statistics and visualizations of known drift patterns, correlations among relevant variables are studied and two main approaches are evaluated. A previously available rule-based method serves as a baseline, while the proposed models include a regression-based method and an Isolation Forest anomaly detection approach.

Both approaches considered achieve 100% recall for drifting-tube detection, precision ranges between 81% and 87%, and the median lead days lies between 121 and 125. The overall findings support the use of ensemble learning, combining the strengths of multiple methods, besides the regression-based and Isolation Forest drift detection, also the rule-based baseline can be included. With further tuning and ensemble learning implementation, the project's methods are suitable for deployment in Databricks, where inference can run automatically on newly uploaded field samples and/or on a time schedule together with continuous evaluation of detection performance.

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Certificate of Advanced Studies in Applied Data Science (CAS ADS)

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Contents

1 Introduction

1.1 Context and Product Description

Industrial X-ray sources are critical components utilized across various sectors for non-destructive testing (NDT) and high-precision quality control, such as electric vehicle (EV) and power grid stabilization battery inspection, and from big metallic parts as e.g. in automotive and aerospace industries down to packaging inspection for chips in semiconductor industry. This final project focuses specifically on the MesoFocus 450 kV (MF-450) X-ray source product line (MesoFocus indicates something between Mini- and MicroFocus, which roughly stand for milli- and micrometer range of focal spot). They are running at end-user sites around the world, mainly in Asia and North America. An industrial X-ray source is fundamentally comprised of an X-ray tube (one kind of vacuum tube, also called electron tube) and an X-ray generator (high voltage power supply and control for the X-ray tube).

The MF-450 operates using five distinct focal spots (FS): 65 µm, 100 µm, 250 µm, 350 µm, 450 µm. In [3] more information about the MF product family, incl. the selectable Focal Spots, is found. When an end-user configures the system, they define the required focal spot and the high voltage from anode to cathode (kV), depending of which kind of sample needs to be visualized (material, size, ...) and how small details need to be seen (required resolution). Based on these user inputs, the system references a lookup table (LUT)—derived from the allowable power density on the anode target—to automatically determine the required grid voltage and emission current. To achieve and maintain this required emission current, the system dynamically adjusts the heater current as a control variable.

1.2 Problem Statement

A critical operational issue has emerged within a specific subset of the MF-450 X-ray tubes, a kind of tube emission degradation, characterized by an anomalous drift in the heater current. This phenomenon typically manifests after a cumulative exposure time exceeding 200 hours. To date, this issue has been confirmed in 13 specific tubes (out of 164 available MF-450 tubes in field data, what is only a part of the total number of sold resp. running at end-user sites).

The impacts of this heater current drift are multifaceted and severe:

- Performance Degradation: The system fails to reach the required emission current setpoint, which directly manifests as a visible reduction in image quality for the end-user.
- Tube Lifetime Degradation: Operating the tube at maximum heater current for extended exposure times drastically reduces the overall lifetime of the X-ray tube.
- Financial and Operational Losses: Unscheduled downtime on the end-user side incurs significant, quantifiable monetary costs per hour.
- Commercial Risk: Equipment degradation leads to customer dissatisfaction, creating inefficiencies for internal support teams and presenting a worst-case risk of long-term reputational damage.
- From a data science perspective, diagnosing this issue presents a severe class imbalance problem. Because only 13 tubes have officially exhibited this behavior, standard supervised classification algorithms are unviable. However, by treating individual heater current time-series samples rather than entire tubes as distinct data points, the dataset becomes sufficiently large for statistical modeling (but unbalance remains).

1.3 Goals and Methodology

The primary objective of this final project is to develop a data-driven model capable of detecting heater current drift, using field data from the X-ray sources available in Databricks, more information regarding Databricks is found in APPENDIX A. Ideally, this detection should occur with a significant lead time—meaning the model flags the onset of a drift long before it impacts image quality or causes damage of a tube.

The model's evaluation metrics are prioritized as follows:

- Primary Metric: $Recall = \frac{TP}{TP+FN}$, TP = true positives; FN = false negatives - The model must maximize recall to ensure no drifting tubes are missed.
- Secondary Metric: $Precision = \frac{TP}{TP+FP}$, TP = true positives; FP = false positives - While false positives should be minimized, precision is a lower priority. Generated alerts are routed internally to technical experts for manual verification rather than being sent directly to customers, meaning false alarms (only) incur internal operational costs.
- Third Metric: Lead time - From drift detection to failure (last available heater current sample)

Note: If one would change 2nd and 3rd priority, having higher priority for long lead time than for precision (still recall as 1st priority), in extreme case, the model would detect all instances, having 100% recall, max. lead time (and bad precision, which is only the 3rd priority)

Given the extreme scarcity of positive (drifting) instances, this study treats the problem as an anomaly detection and statistical monitoring task. The core methodology involves establishing a baseline of "healthy" system behavior using historical data and evaluating deviations from this norm. To achieve this, two distinct data science frameworks will be implemented, evaluated, and compared (also to a rule-based approach, which was being developed outside of this project, which is used as a baseline):

1. Regression-Based Drift Detection (Residual Analysis): A predictive model is trained to estimate expected heater current under normal conditions. By continuously calculating and evaluating the residual (the difference between the actual observed heater current and the model's prediction), statistical drift can be detected.
2. Isolation Forest-Based Drift Detection: An unsupervised tree-based ensemble algorithm will be utilized to explicitly isolate anomalies rather than profiling normal data points. Because drifting samples deviate significantly from typical operational patterns, they require fewer splits to be isolated within the tree structure, assigning them a high anomaly score.

During the inference phase, both approaches will be rigorously tested on a combined dataset of healthy and degraded tubes to determine their effectiveness in catching the drift early.

2 Data

2.1 Data Flow

Data processing is organized according to a multi-layer architecture called Lakehouse Architecture with the layers Bronze-Silver-Gold (medallion), which is best practice in Databricks [7]. More about the Databricks platform can be read in APPENDIX A, incl. related links. The following figure shows a simplified overview of data flow from data source to analysis, plotting, model training and evaluation in Python notebooks:

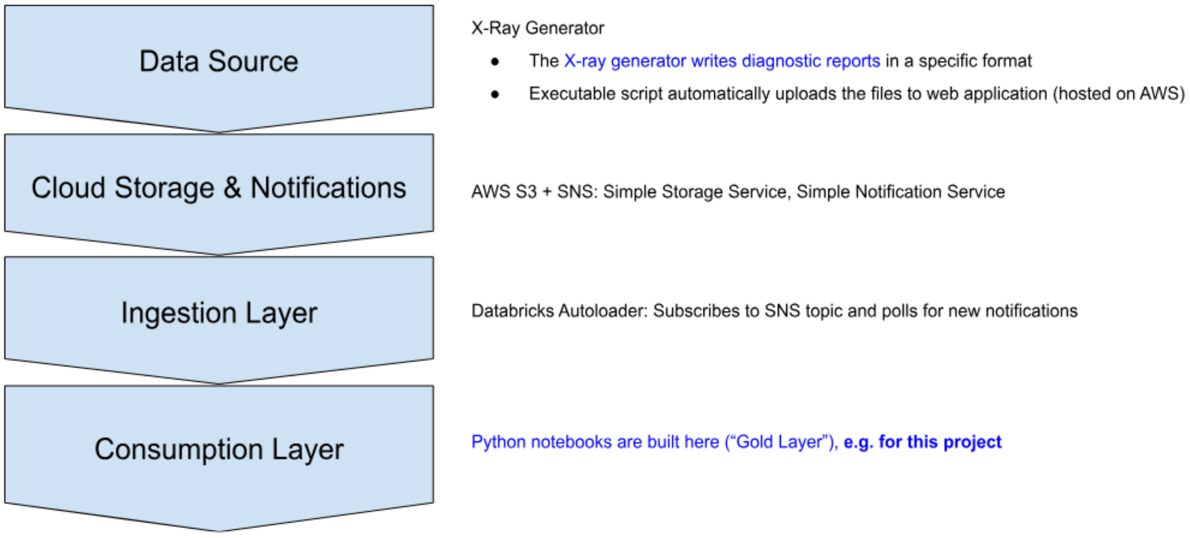

Figure 1: Data flow diagram, from data source to Python notebooks

2.2 Metadata

As one part of metadata, date and time of script execution is recorded in Databricks. Therewith, the reproducibility of analysis is given, since the data changes and additional entries are written into the table(s) automatically with a timestamp. Freezing a copy of the required tables which are being used for analysis would also be possible.

Metadata for field data itself, are the units of physical values in the table, e.g. high voltage in kilo volts (kV), emission current in milli-Amperes (mA), durations in seconds (s), etc.

Another kind of metadata would be the customer and location of end-user application. For data security reasons, this information is separated and not accessible from Databricks, but only from other platforms. For the project here it should not be relevant, the data from the diagnostic reports and therewith available in Databricks should be sufficient and therewith no security issue is expected.

2.3 Data Quality

The accuracy and precision per entry to be expected in the available field data should be more than enough, since the values come from the device (X-ray generator) internal measurements which are also used for tracking the setpoints given by the user, where high accuracy and precision is required (electronic component tolerances in measurement chains, analog-to-digital conversion number of bits, input signal range etc. is chosen accordingly in product design).

Regarding completeness it is important to mention, that the data uploaded and accessible from Databricks analysis is biased: Firstly, not all end-users want or are allowed to upload diagnostic reports, e.g. because of confidentiality (e.g. no data is allowed to leave the fab), or secondly, those end-users which allow to upload data, do it mostly for devices and tubes having a problem or even a defect, not for the devices which are running smoothly (herewith, accessible data is expected to have a "pessimistic" tendency in some regard, at least towards the end).

A risk linked to the fact that not from all devices running in the field data is uploaded: Drifts during ‘blindspots’ will never be detected, what means when there is no upload before the drift is leading to negative, undesired consequences, obviously there is no chance to react beforehand.

2.4 Data Model

The data model at the conceptual level, the logical level and the physical level is explained just briefly, the data model design is not part of this project, since the data is already available on Databricks consumption layer, ready to work with.

Conceptual Data Model

On the conceptual level, there are entries per row in the following form: Timestamp (date + time), device ID (generator), tube ID, tube type, event type (exposure, warm-up, work-point change, arc, ...) incl. duration, operating point, consisting of high voltage in kV, focal spot, emission current in mA, etc., temperature, measured at several locations in the generator device, and more

Logical Data Model

The names (incl. data type) of all the 45 columns from "svc_datacore.external.silver_blox_enriched" are:

device_id:string, tube_model_id:string, start_time:timestamp, end_time:timestamp, duration_seconds:double, duration_hours:double, event_type:string, shutdown_type:string, reason:string, cycle:double, wpch_count:long, wpch_within_expo:boolean, arc_count:long, hv_kv_set:integer, tucu_ma_set:double, filcu_a:double, grid_voltage:double, pwr_w_reached:double, focal_spot:integer, tube_max_w:double, mid_exposure_focal_spot_change:boolean, comp_a_tank:string, comp_c_tank:string, comp_software:string, comp_ifc:string, comp_ecu:string, comp_poc1:string, comp_poc2:string, temp_c_tank:double, temp_a_tank:double, temp_ecu_oil:double, temp_fgu_oil:double, temp_poc1:double, temp_poc2:double, temp_ifc:double, tube_id:string, tube_model:string, tube_family:string, tube_category:string, tube_cycling:boolean, device_model:string, polarity:string, setpoint:string, agg_duration_hrs:double, created_at:timestamp

Physical Data Model

The dataset is available in Databricks where also the compute clusters for data analysis, model training and evaluation etc. are. “silver_blox_enriched” data table size and dimension:

- 709 MiB (Mebibyte: 1 MiB = 2^{20} Bytes = 1,048,576 Bytes)
- rows x columns: 37’940’853 x 45, this number of rows is the current one (2026-06-30) and is growing over time with new entries from the field devices uploads.

2.5 Data Pre-Processing and Data Cleaning

Device-tube pairs with specific tube type MF-450 are selected for further processing. The data is filtered regarding event type to only work with heater current samples during events “Exposure”, “Warm-Up” and “Disrupted Warm-Up”. Besides the target variable ’heater current’, focal spot, high voltage, emission current, power and grid voltage are used as features. All entries containing missing or NULL values (NaN, None) are dropped. The positives, what here are the damaged device-tube pairs, are selected “by hand” and made available for drift performance evaluation. A train-test-split is performed with a ratio of 80:20 (test set: all damaged + random healthy device-tube pairs to fill up to 20%; train set: the remaining healthy device-tube pairs).

3 Exploratory Data Analysis

3.1 Descriptive Statistics

The following table gives an overview regarding statistics for number of total and specifically MF-450 device-tube pairs in the field, thereof damaged ones showing heater current drift: Dataset dimension, number of heater current samples and number of pairs.

description	value
total dataset dimension: rows x columns	37'940'853 x 45
MF-450 dataset dimension: rows x columns	290'122 x 45
damaged pairs dataset: rows x columns	119'758 x 45
total dataset: number of heater current samples	11'840'069
MF-450 dataset: number of heater current samples	289'975
damaged pairs dataset: number of heater current samples	49'308
total dataset: number of device-tube pairs	1'587
MF-450 dataset: number of device-tube pairs	164
damaged pairs dataset: number of device-tube pairs	13

Table 1: Statistics of dataset dimension, number of heater current samples, and number of device-tube pairs

Below a summary statistics regarding number of heater current samples per pair is shown, separately for all pairs, damaged pairs and healthy pairs:

description	all pairs	damaged pairs (drift confirmed: positives)	healthy pairs: negatives
minimum	1	478	1
maximum	30'008	16'820	30'008
mean	1'768.14	3'792.92	1'593.82
median	232	2'020	170
standard deviation std	4'336.46	4'604.46	4'284.00

Table 2: Statistics of heater current samples per pair

3.2 Time-Series Data Visualization

So far, 13 tubes are identified having the heater current drift, what means confirmed by 3 different persons having domain specific knowledge had a look at the data resp. plots and confirmed this kind of anomaly resp. drift.

Here, the plot of only one of the 13 known drift instances is shown, the complete set of drift plots is found in APPENDIX B. For indicating the operating point dependence of Heater Current, different markers are used per end-user defined setpoint {Focal Spot, High Voltage}.

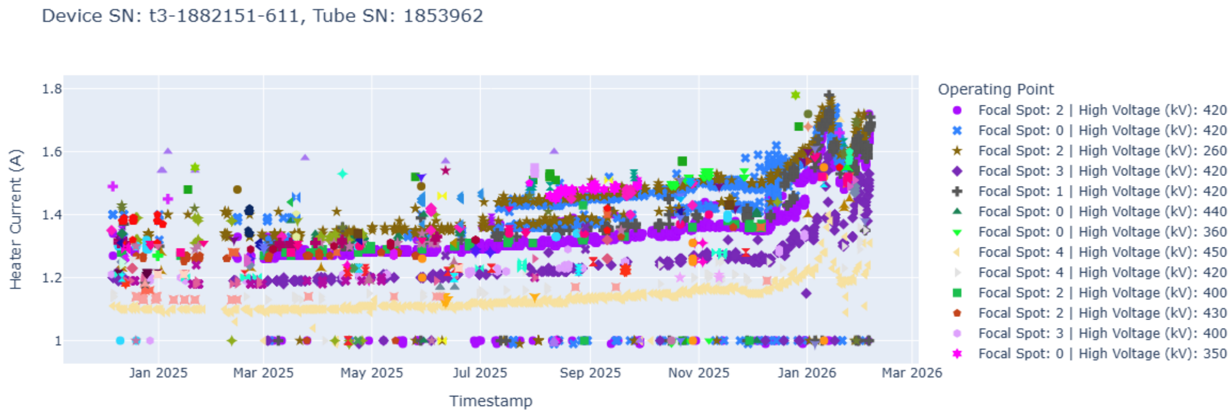

Figure 2: Drift time-series example: Heater current samples, different marker per operating point

When visually analyzing the plots of all 13 drifts, which all look differently, it can be observed, that a process responsible for the drift, is sometimes also continued during breaks, see heater current steps before to after the breaks => indication that it does not come from exposure time ("drift increases with more exposure time"), but also during inactive time (process to be defined).

3.3 Correlation Analysis Visualization

Below the correlations between the chosen columns heater current, focal spot, high voltage, emission current, power and grid voltage (those are all signals resp. variables directly related to the X-ray tube available in the data and therefore taken as a starting point, also for model training as shown in Chapter 4), from MF-450 dataset, are visualized in a correlation heat map.

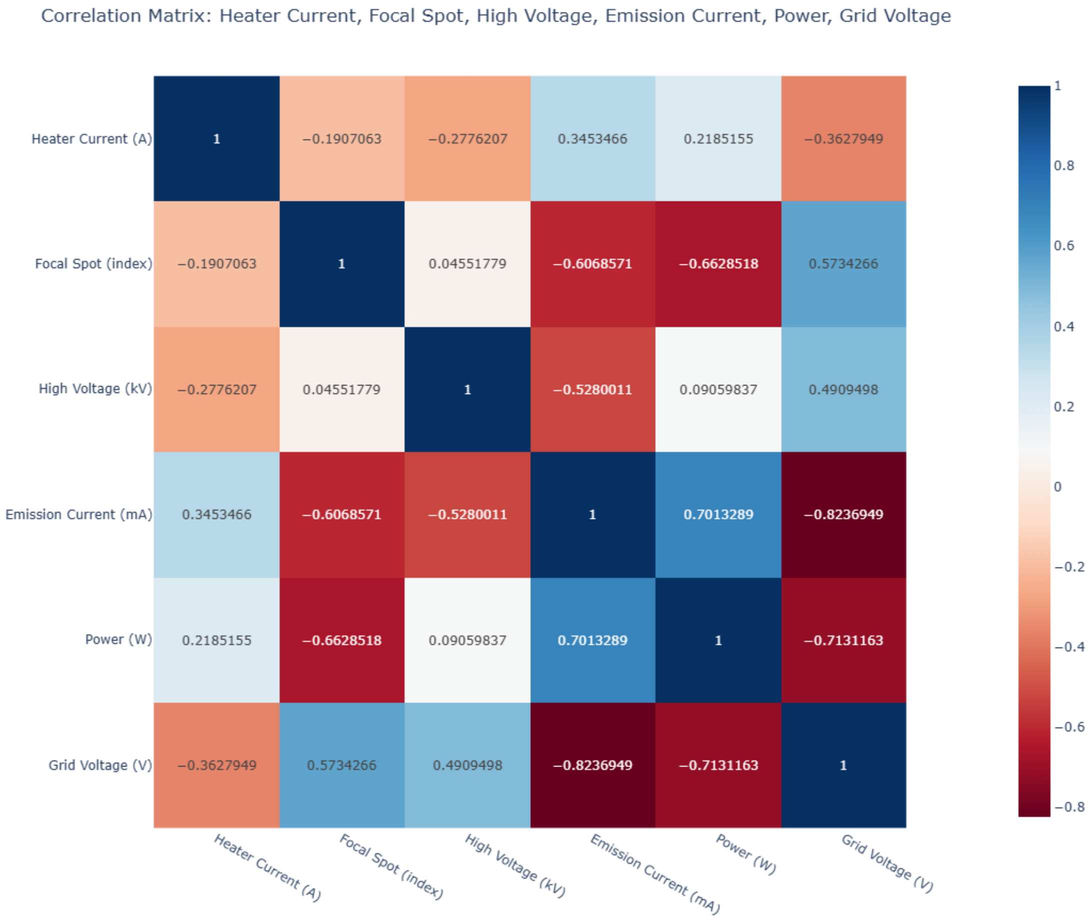

Figure 3: Correlation heatmap: Correlations between chosen columns from MF-450 dataset

Some of the correlation coefficients between the variables do not have values close to ± 1, but closer to ± 0.5, what can be explained by the non-linear relations e.g. between emission current (current in tube from anode to cathode) and anode to cathode voltage (high voltage), for more information regarding vaccum tube and specifically X-ray tube characteristics, e.g. Child-Langmuir law, see [1]. It is also interesting to see the relations with low correlation, coefficients close to 0, which fit to the parameters independently to each other, set by the end-user, high voltage and focal spot, or high voltage to power (as power is derived from the LUT and depending from the focal spot, power density on anode/target).

4 Approach and Methods

Building upon Goals and Methodology outlined in Section 1.3, this chapter details the approaches and methods considered to detect heater current drift. To establish a benchmark for performance, a traditional rule-based drift detection algorithm is implemented as a baseline. This baseline is then compared against two distinct data-driven approaches: A regression-based model using residual analysis, and an unsupervised Isolation Forest model. Together, these methodologies explore the spectrum from rigid, threshold-based monitoring to advanced machine learning anomaly detection. During concept phase, also other potential methods were considered (Local Outlier Factor LOF, One-class SVM with non-linear kernel RBF, Neural Networks based, e.g. autoencoder and use of reconstruction error to detect anomalies), but for this final project work with limited time budget, the two methodologies regression-based and Isolation Forest are enough. If the two chosen methodologies besides rule-based are performing satisfactory, this would allow to use three approaches orthogonal to each other combining to a more robust setup, see 4.4 Ensemble Learning.

What has to be mentioned here (compare to 2.4 Physical Data Model), is that the dataset changes over time, as new data is uploaded, and therewith also the results will change when executing the scripts with new data. All what is shown in this report, is based on data status and execution on 2026-06-30.

4.1 Baseline: Rule-based Drift Detection

As already mentioned, the rule-based drift detection serves as a baseline for the machine learning algorithms considered during this project, and it was designed and implemented outside this CAS project and not by the author.

Approach

This method relies on static, "hand-crafted" limits designed around all of the known positive (drifting) cases from historical operating data. The algorithm monitors the heater current, triggering an alert when there is an increase in the current value from one sliding window to the next. A limitation of this approach is that it must be evaluated for each operational setpoint separately. To optimize the algorithm, a grid search is performed using the following hyper-parameters:

- Baseline hours: The total exposure time, including warm-ups and disrupted warm-ups
- Threshold percentage: The fixed limit used to trigger an alert when exceeded
- Sliding window size: The number of heater current samples analyzed in one window

While this approach is relatively simple to implement and highly transparent, it does not perform generalization from a training set to a test set. This lack of generalization contradicts a basic principle of machine learning, where a system learns broader patterns from specific examples—a concept discussed in detail, e.g. in [2].

Result

A short summary of results is given here, what is enough to compare to the other two approaches considered during this project.

Performance using the selected combination of hyper-params after grid search - Window size 5, threshold 6%, baseline hours 10:

- True positives: 13
- True negatives: 43
- False positives: 9
- False negatives: 0

This gives a Recall = 100% and a Precision = 59%. The lead time from detection to failure (last heater current sample available) is 103 calendar days median and 169 days mean.

4.2 Regression-based Drift Detection

This approach uses supervised machine learning to learn what a "healthy" heater current should be under normal conditions. The specific X-ray tube degradation manifests as an increase in heater current at a given operating point. A regression model is trained on healthy device-tube pairs to predict the expected heater current ('target variable') from operating parameters — focal spot, high voltage, emission current, power, and grid voltage. Three algorithms are compared: Linear Regression, Random Forest, and XGBoost ([8], [9] and [10] provide documentation and Python interface description for the three algorithm implementations). During testing, the model's prediction is comapred to the actual measured heater current; the signed difference between the measured and predicted heater current is called the residual. For a healthy tube the residual remains near zero; for a degrading tube it trends positive, flagging an anomaly resp. drift.

Approach

As introduced above, three model architectures are compared: Linear Regression, Random Forest, and XGBoost. Hyper-parameters for the latter two are selected by grid search minimizing mean squared error. The model with the lowest test-set MSE is carried forward.

Raw residuals are not directly comparable across operating conditions or device-tube pairs. Two normalization steps are applied. First, residuals are z-scored within operating groups defined by focal spot and high voltage bin, using statistics computed from the training set. This removes the effect of operating point differences on residual magnitude. Second, the median of the first N residuals for each device-tube pair is subtracted, removing the inter-tube differences so that drift appears as a deviation from each pair's own healthy baseline. An experimental improvement replaces the median subtraction with an OLS (Ordinary Least Squares) line fitted through the first N residuals, additionally removing any initial slope.

Drift is detected via three signals computed on the normalized, baseline-adjusted residual time series. The primary signal is a one-sided CUSUM (Cumulative Sum), which accumulates evidence of a sustained positive shift and alarms when the cumulative sum exceeds a threshold h. A rolling OLS slope and a rolling signed mean serve as complementary signals that respond more quickly to step changes and slow monotone trends respectively. The three signals are fused — configurable as OR, MAJORITY, or AND (finally, OR is used, to be maximal sensitive, resp. to achieve max. recall) — and a minimum number of consecutive alarm samples is required before a device-tube pair level detection is confirmed. Detection thresholds are selected by grid search on the test set, optimizing for 100% recall on confirmed damaged pairs, minimum false positives, and maximum detection lead time.

Results

To check the relevance of the 5 selected features, the feature importance plots for LR, RF and XGB are created (it was also done using less features, incl. performance evaluation, due to best performance with all 5 features, in this report everything is shown with 5 features).

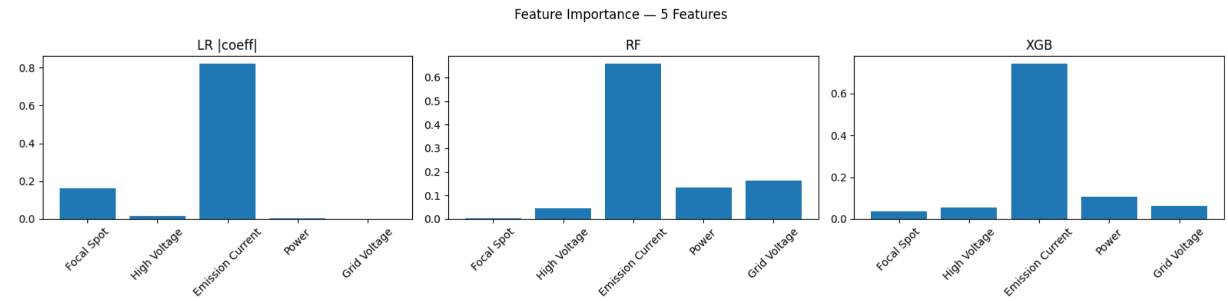

Figure 4: Feature Importance with 5 Features, heater current as target variable, for Linear Regression (LR), Random Forest (RF) and XGBoost (XGB)

The table below gives an overview regarding regression models performance (5 features): Mean absolute error (MAE), mean squared error (MSE) and R-squared. For Random Forest and XG-Boost the results are shown using the best hyper-parameter set found by grid search.

Model	Split	MAE	MSE	R^2
Linear Regression (LR)	Train	0.06071	0.00674	0.5579
	Test	0.07983	0.01125	0.5021
Random Forest (RF)	Train	0.01228	0.00062	0.959
	Test	0.08742	0.01201	0.4686
XGBoost (XGB)	Train	0.02908	0.00288	0.8108
	Test	**0.07201**	**0.01002**	**0.5565**

Table 3: Overview of 5-feature regression models performance: LR, RF and XGB compared

XGBoost shows best performance on test dataset and is therefore used for residual-based drift detection. Residual-based signal evaluation parameters are also found by grid search. The following plot shows as an example evaluation of the same positive example as already seen in 3.2 and how the alarm is triggered by the thresholds on evaluation signals

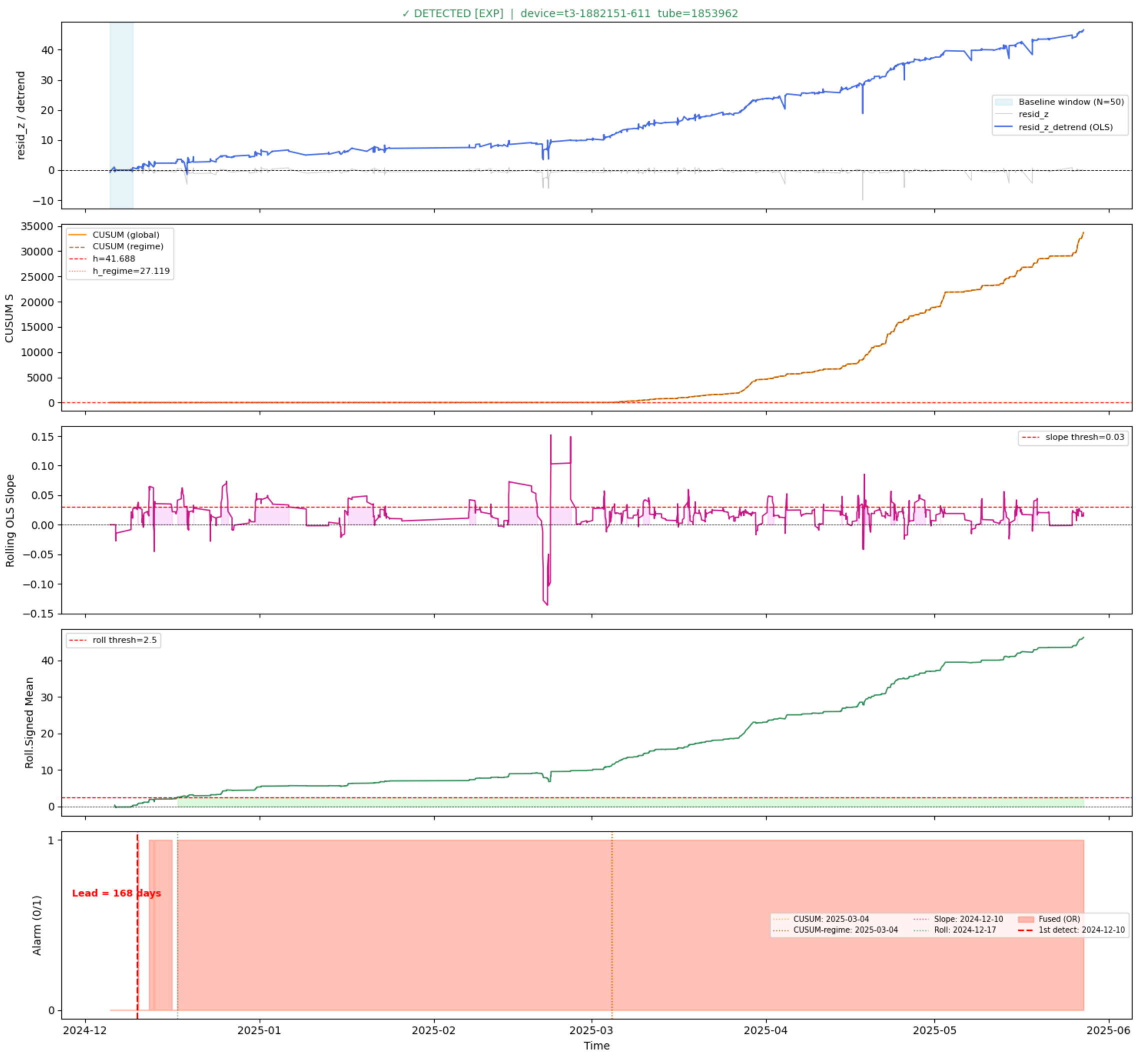

Figure 5: Evaluation of residuals from XGBoost (5 features) incl. alarm trigger visualization

Using this best hyper-parameter set found for regression model XGB, this leads to a recall =

100% and precision = 81%. See the confusion matrix below:

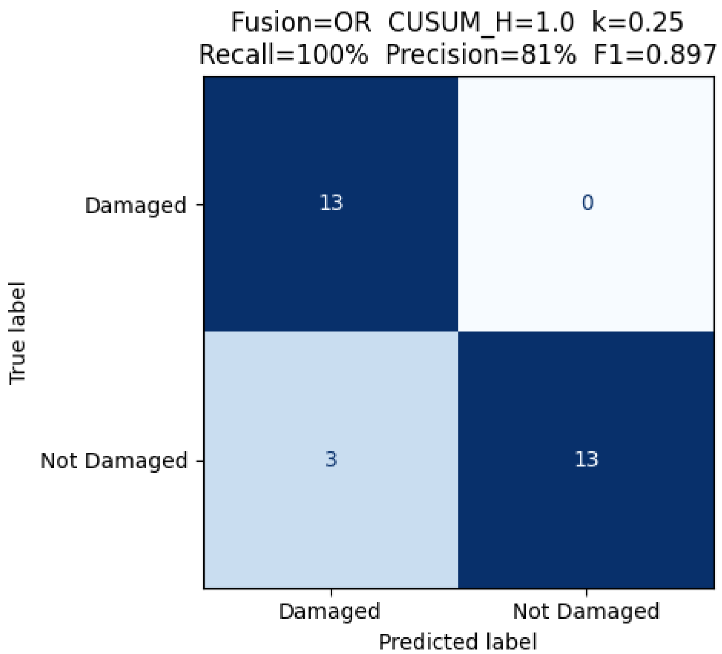

Figure 6: Confusion Matrix for XGBoost regression-based drift detection (5 features)

The 3rd priority, after recall and precision, is the detection lead time, what is the time from detection to failure (last heater current sample). The following plot shows the lead time for all 13 positives:

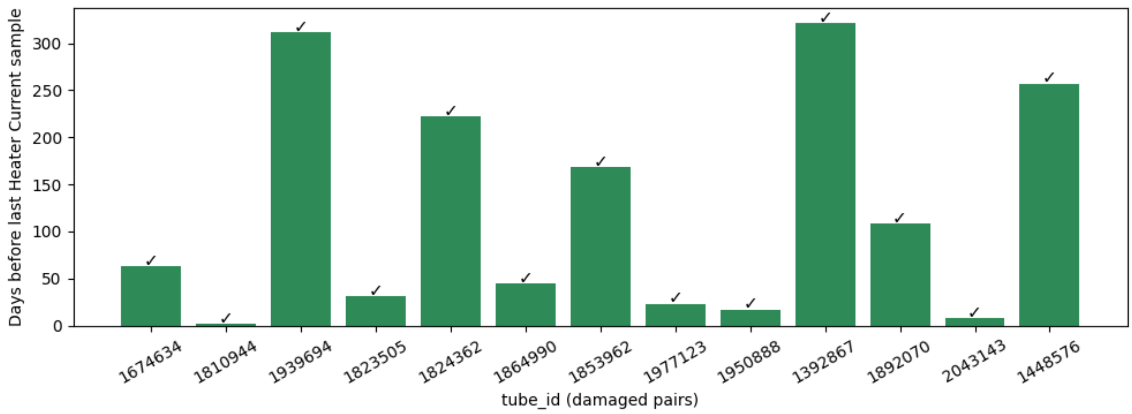

Figure 7: Detection lead time (from detection to last heater current sample) for XGBoost-based drift detection, 5 features

This gives an overall lead time median 63 calendar days, mean 121 days, based on the 13 positives available.

These results for recall, precision and lead time are somehow impressing, when remembering the ("not so high") $R^2 = 0.56$ on the test set.

4.3 IsolationForest-based Drift Detection

The second data-driven approach uses the Isolation Forest algorithm, which takes an unsupervised approach to anomaly detection (see [11] for documentation and Python interface description for the Isolation Forest algorithm implementation). Instead of learning what is normal, this algorithm actively tries to isolate individual data points using a network of random decision trees. Therefore, it requires no labelled failure examples — only a reference population of healthy tubes. This makes it well suited to the present problem, where confirmed failures are rare (13 out of 164 MF-450 device-tube pairs where field data is available) and the damaged-tube distribution cannot be characterized reliably. Because drifting data points have unusual, abnormal values, they stand out from the rest of the data and require very few steps to be isolated. The algorithm converts this ease of isolation into an "anomaly score", allowing us to catch drifting tubes directly without needing to predict a specific target value.

Approach

The algorithm builds an ensemble of random binary decision trees. Each tree recursively splits a random data sub-sample using randomly chosen features and split values. An observation that is easy to isolate — requiring few splits to separate from the rest — receives a high anomaly score. Observations embedded in dense, normal regions require many splits and receive low anomaly scores. The model is trained on healthy tube pairs only. At inference time, observations from a degrading tube draw anomalously high heater current relative to the operating point and are isolated quickly, producing high anomaly scores that trigger an alarm.

Like the regression approach, the method must account for tube-to-tube variance. A single global alarm threshold is used.

Also here, hyper-parameters are selected by grid search: number of trees, samples per tree, contamination fraction, rolling smoothing window for heater current. The optimization objective is maximum recall on confirmed damaged pairs, followed by minimum false positives and maximum detection lead time.

Results

The plot below shows the heater current (raw and smooth), the anomaly score (IF score) and the alarm trigger for the best hyper-parameter set found by grid search, again for the same example as seen already in chapters 3.2 and 4.2.

Figure 8: Evaluation of 'IF Score' from Isolation Forest (IF, 5 features) incl. alarm trigger visualization ("Drift flagged")

With the best hyper-parameter set found for Isolation Forest IF, the recall is 100% and precision 87%, what is slightly better than the XGB regression model combined with residual analysis.

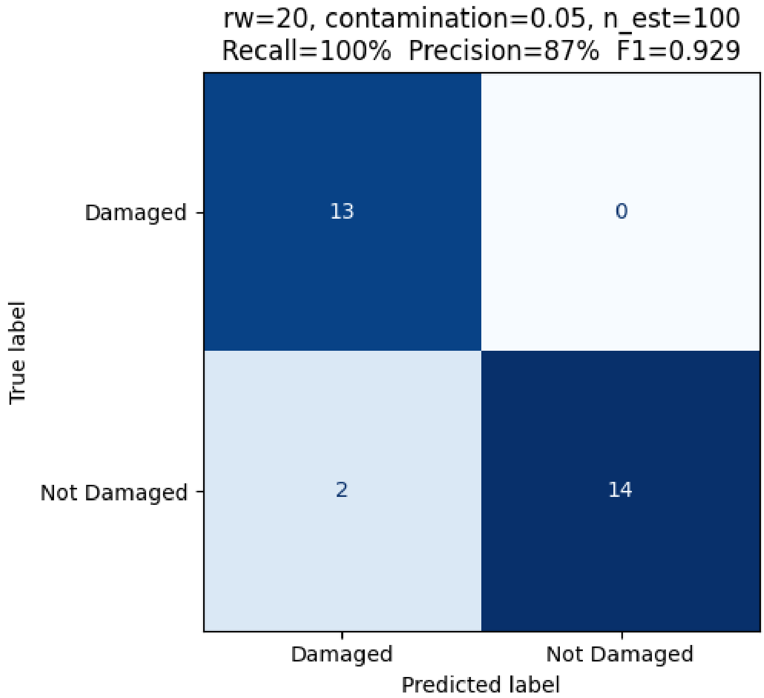

Figure 9: Confusion Matrix for Isolation Forest (5 features)

Below the detection lead time for all 13 positives is shown:

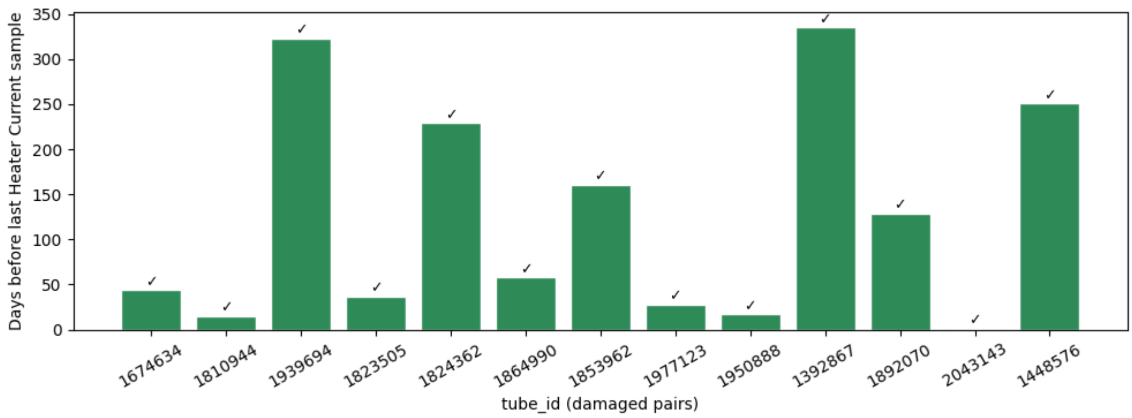

Figure 10: Detection lead time (from detection to last heater current sample) for Isolation Forest, 5 features

This gives an overall lead time median 58 calendar days and mean 125 days, based on the 13 positives available in the dataset.

Note: When prioritizing the lead time higher (and allowing the precision to decrease), still recall 100%, this would lead to 4 FP instead 2 FP (Precision only 76% instead 87%), but median lead=120 days, mean lead=138 days. This was done only on an experimental level for Isolation Forest, to have a look what happens with the lead time if more false positives are tolerated. This trade-off is discussed more in detail below in Chapter 5 "Results Discussion".

4.4 Combination of Different Approaches: Ensemble Learning

To maximize robustness, this approach explores ensemble learning by combining the individual strengths of the previously discussed models, being orthogonal to each other, more on this can be found in [2]. This framework evaluates techniques such as Stacking and Voting classifiers, specifically focusing on a logical "OR" union operation that merges the outputs of the rule-based baseline, the regression model(s), and the Isolation Forest working in parallel on the same dataset. Under this ensemble strategy, an alert is triggered if any single model flags a data point as an anomaly. This collective approach guarantees high sensitivity and increases the chance that no drifting tubes are missed, directly in alignment with the high-recall as 1st priority requirement of the project. This can be described in such a form (a simple one to start with, it can also get more sophisticated if meaningful or needed), where rule-based (RB), Regression (Reg) and Isolation Forest (IF) scores are combined by logical union to get the ensemble learning result $\mathrm{Y}_{\mathrm{ensemble}}$:

$$Y_{ensemble} = RB_{score} \vee Reg_{score} \vee IF_{score}$$

5 Results Discussion

In the table below it is shown, that the results of approaches analyzed additionally to rule-based method are performing on a comparable level to rule-based approach (Recall, precision, lead time). As all three methods reach recall 100%, what was the 1st priority, all three perform at least satisfying. Therefore, all three approaches are interesting to be considered, they do not have to compete, but could be combined in an ensemble learning approach as introduced above.

Approach	Recall (%)	Precision (%)	Median lead time (days)	Mean lead time (days)
Rule-based	100	59	103	169
Regression (XGB)	100	81	63	121
Isolation Forest (IF)	100	87	58	125

Table 4: Performance overview for Rule-, Regression- and IF-based drift detection

What needs to be repeated here, is that the rule-based algorithm was evaluated on the whole dataset after finding the best hyper-parameter set, found by grid search, also based on the whole dataset (no train-test split as it is done for the machine learning approaches), and therefore the results can not be compared directly to the Regression and Isolation Forest drift detection results.

As introduced already at the end of Chapter 4.3, when the 1st priority metric has reached the maximum, recall = 100%, there is the trade-off between precision and lead time to detection (resp. 2nd vs 3rd priority in hyper-parameter optimization). During this project, precision and not lead time was taken as 2nd priority, because otherwise the algorithm could just classify at the from the beginning all instances to positive and therefore get 100% recall and maximal (best possible) lead time, but having the worst precision what is “only” 3rd priority. That is also the reason, why in the table above regression and IF-based detection have better precision than the rule-based method (where the prioritization and hyper-parameter selection was done differently), but the lead time is shorter (worse) compared to the rule-based approach. What would be possible with the machine learning approaches, is to still give 1st priority to recall, 2nd priority to lead time, but to specify a boundary condition for precision somehow, what would be more complicated due to the need of additional definitions.

Regarding ensemble learning, an idea is also to specifically combine a dedicated mixture of models, where all of them still are optimized for maximum recall as 1st priority, but as 2nd priority a part of them is optimized for precision and the other part as 2nd priority for lead time (while still keeping a min. precision t.b.d.). This would be in addition to having a combination of different kinds of algorithms being orthogonal to each other, for example as the ones considered in this project, rule-based, regression-based and Isolation Forest.

6 Conclusion and Outlook

The results show, that all three approaches, rule-based, regression- and IF-based perform at maximum 1st priority metric, all having recall = 100%. Regarding precision and lead time, all three models give satisfying results in the range 59% to 87% resp. 58 to 103 days median lead time from detection to damage. When recall has reached the maximum value of 100%, there is the trade-off between precision and lead time.

What the author experienced, is although drift detection using regression is easier to understand than anomaly detection using Isolation Forest, Isolation Forest was able to achieve better results with less effort and less time invested compared to regression, which takes a roundabout approach via residuals and their processing. This speaks for the anomaly detection specific algorithms to use for drift detection, they seem to be well suited, at least Isolation Forest which was tried out during this project.

Due to few positives, better comparability to rule-based approach and limited time, the train-validation-test split, to test after the hyper-parameters are defined, without ever having seen the test data before (hyper-parameters would be optimized during validation step), was not done. Instead, only a train-test split was performed. Therefore, the real test phase will start from now on, with each new sample uploaded (or it could be done still after report hand-in, with the data available, to check the performance using the train-val-test split). At least it is shown, that there are machine learning based approaches, which perform at least on the same level as rule-based, and they could be combined for ensemble learning.

After some fine-tuning, improvements and the ensemble learning implemented, ideally with the idea written down at the end of Chapter 5 (mixture of 1. recall + 2. precision and 1. recall + 2. lead time optimized models), is to embed the algorithms into Databricks, where inference execution is triggered automatically with each new sample uploaded from the field, and/or time step dependent, incl. drift detection evaluation.

7 Acknowledgments

I would like to express my special thanks of gratitude to

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- my employer Comet Group, Industrial X-ray Modules Division and corresponding senior management team members, for the sponsorship;
- Alexandra Schulz, data project manager, for the discussions about business relevant use cases and delivering the required information regarding dataflow and the rule-based drift detection implementation used as a baseline here;
- my family for the support and giving me the freedom of investing the hours, in addition to my daily work.

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List of Figures

8 APPENDIX A - Databricks Platform and Infrastructure

For this project, the Databricks infrastructure is used, since the existing field data of our corresponding Industrial X-ray generator platform is already ingested there and it offers a wide range of well-suited functionality. Databricks is a unified cloud platform for efficient data analysis and machine learning [4]. It handles large data volumes (e.g. as here log files resp. diagnostic reports). In Databricks, the analysis and model building can be done using Python notebooks, dashboards and SQL queries, or a combination of them.

Databricks is built on top of Apache Spark [5]. Data is typically handled using PySpark DataFrames [6], which support efficient, paralleled operations on large datasets that exceed single-machine memory limits. For smaller or exploratory workloads, these PySpark DataFrames can be converted to Pandas DataFrames, allowing more flexible local analysis with familiar Python libraries. This hybrid workflow combines Spark's scalability with Pandas' ease of use for targeted data exploration and visualization.

Within Databricks, Spark enables distributed data processing across scalable clusters of compute resources. Personal compute clusters can be configured, what was done for this project by the author: Databricks runtime 18.1 ML, Apache Spark 4.1.0, Scala 2.13; Preferred node type: 16 GB Memory, 4 Cores.

Databricks bills on a "pay-as-you-go" model, using processing power per hour" and different service tier levels.

9 APPENDIX B - Drift Time-Series Plots

A time-series plot is shown for all 13 device-tube pairs where the drift is confirmed by today.

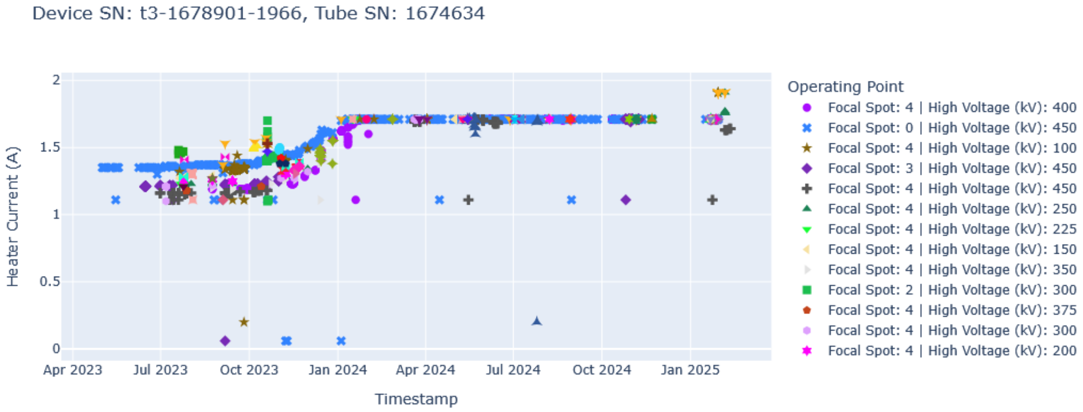

Figure 11: Time-series plot for #1 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

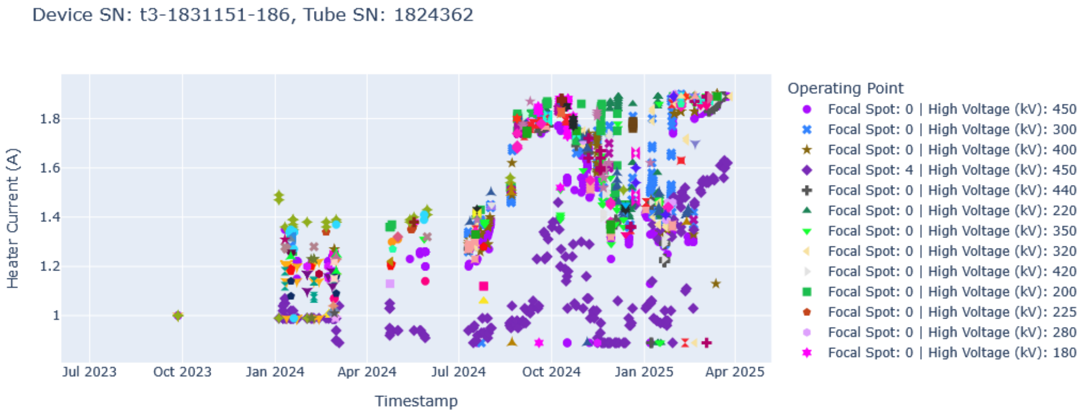

Figure 12: Time-series plot for #2 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

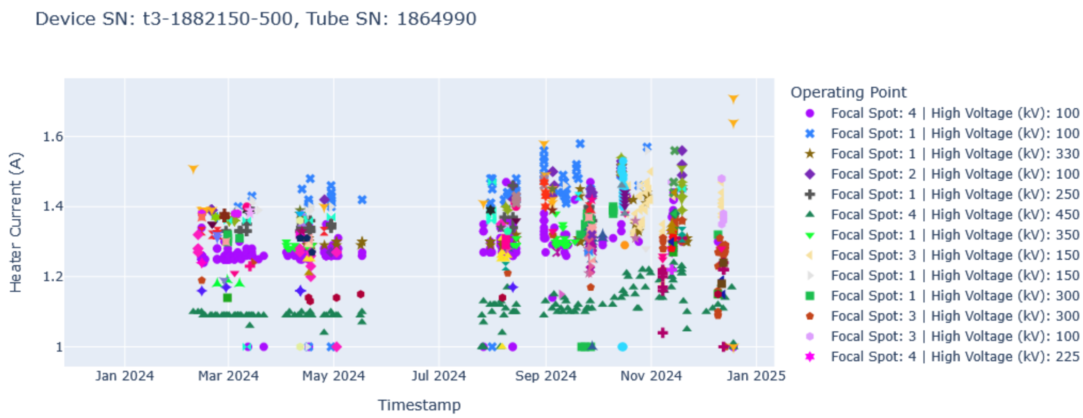

Figure 13: Time-series plot for #3 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

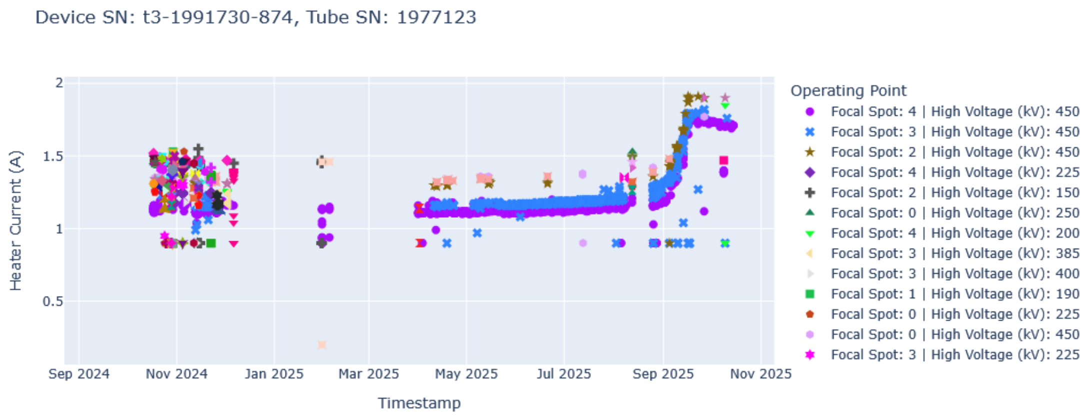

Figure 14: Time-series plot for #4 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

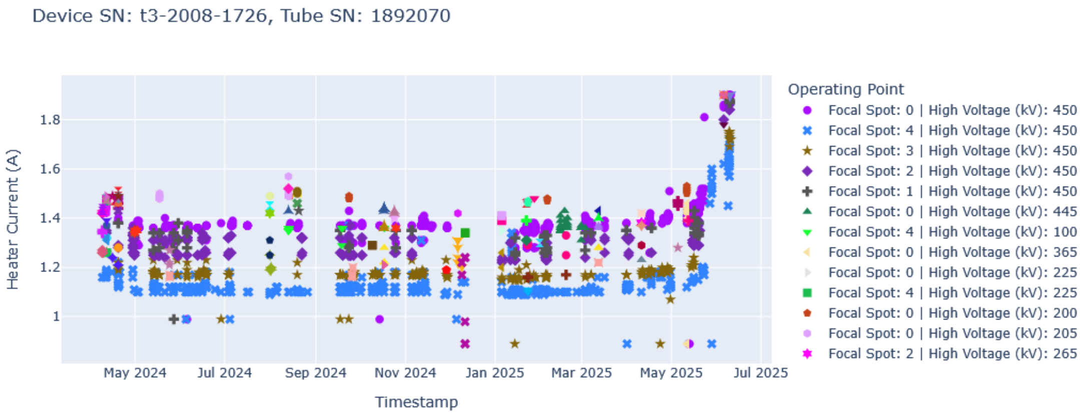

Figure 15: Time-series plot for #5 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

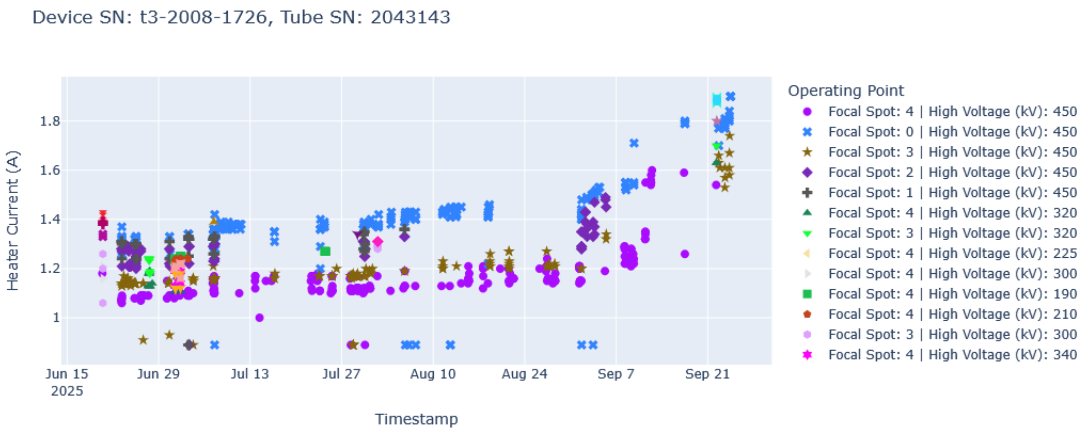

Figure 16: Time-series plot for #6 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

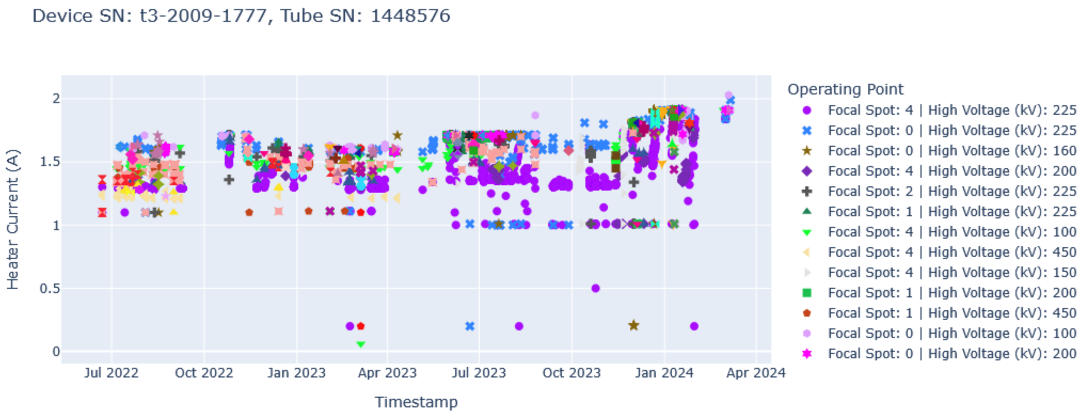

Figure 17: Time-series plot for #7 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

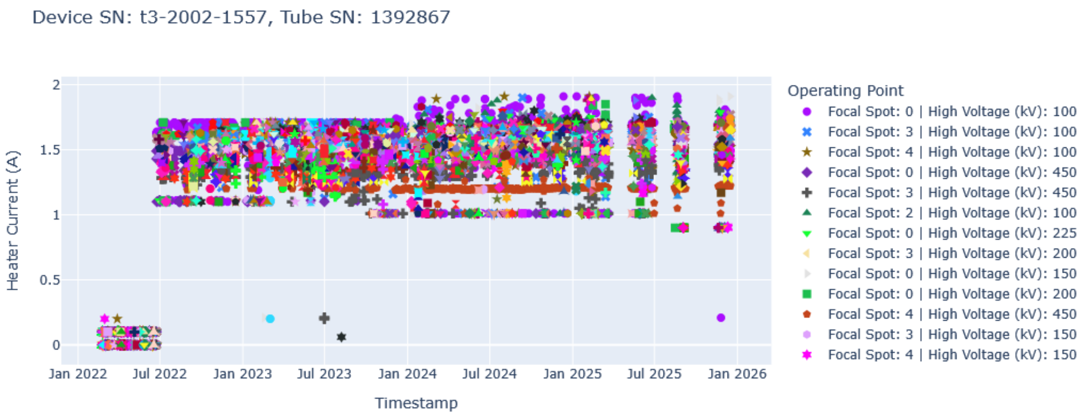

Figure 18: Time-series plot for #8 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

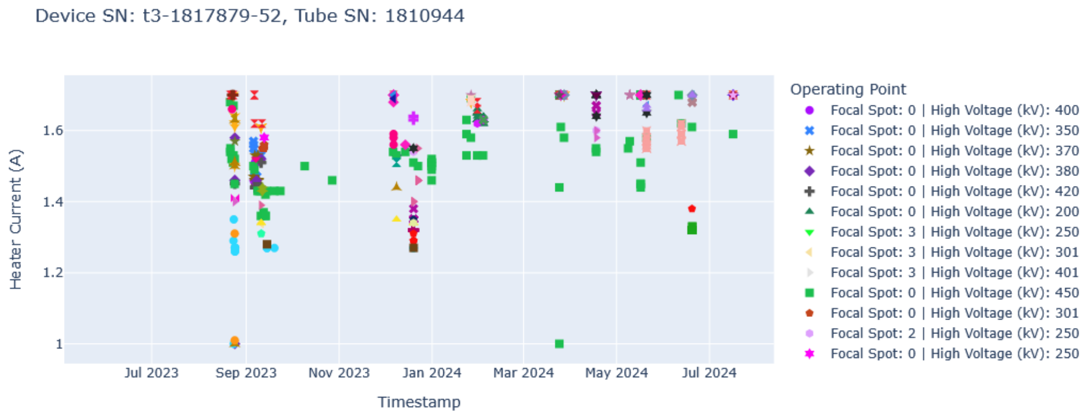

Figure 19: Time-series plot for #9 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

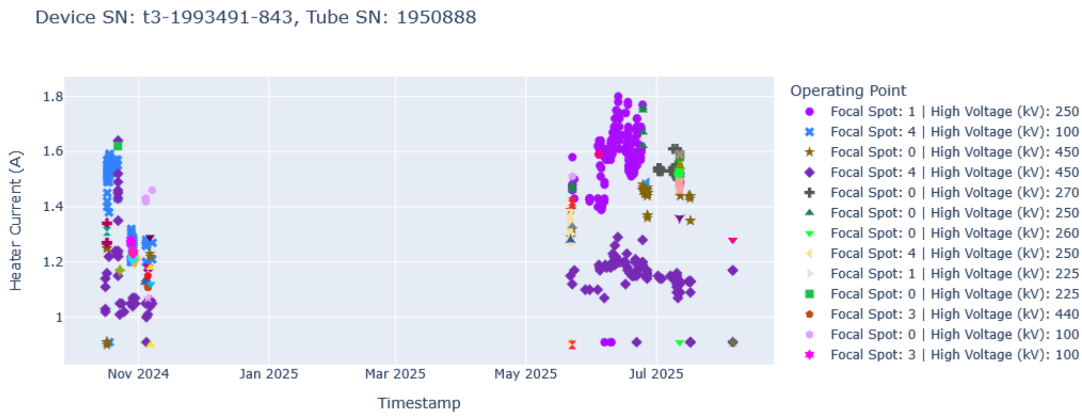

Figure 20: Time-series plot for #10 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

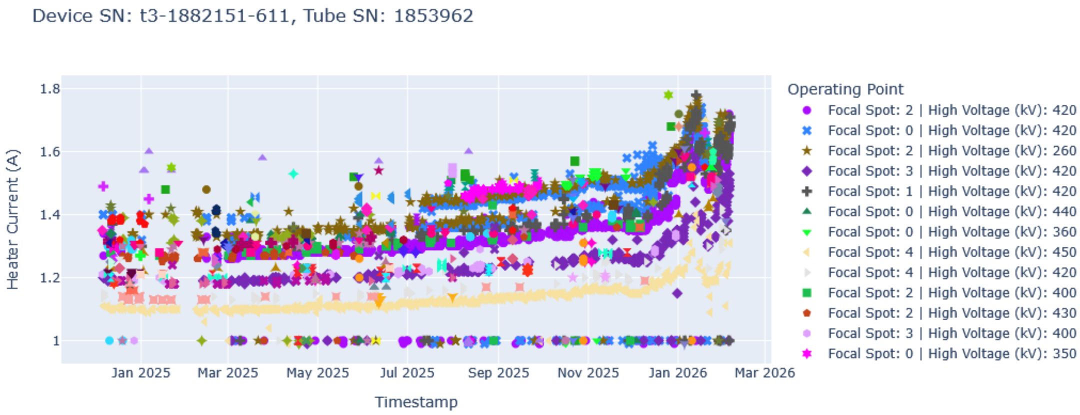

Figure 21: Time-series plot for #11 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

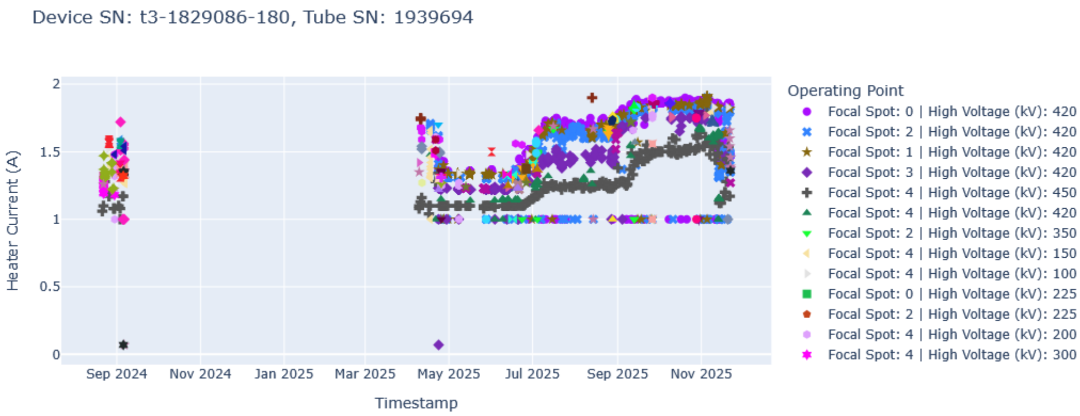

Figure 22: Time-series plot for #12 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

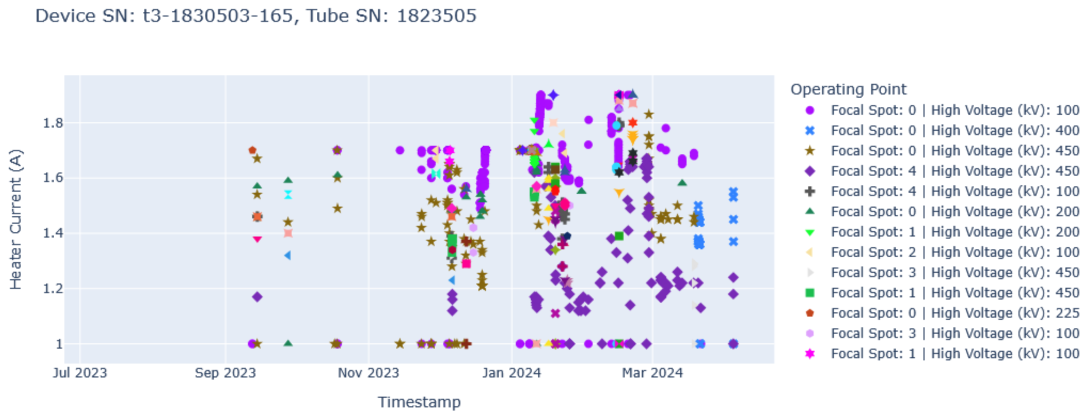

Figure 23: Time-series plot for #13 out of 13 device-tube pairs where the drift is confirmed: Heater Current samples, different marking per operating point

10 APPENDIX C - Artificial Intelligence Assistance

- Generative AI tools (chatbots only, no Agents as e.g. Claude Code or GitHub Copilot) were used for generating more and better Python code in the time available. The ideas, concepts and direction what exactly to implement completely came from the author

- The text in this report was composed and written by the author. In some places, the DeepL translation tool was used to translate from the author's native language, German, into English. For some other text passages, generative AI was used to generate text from bullet points giving content and wording, which afterwards was written by the author in own words.